

$$P := \{p \in \mathbb{N} : p \text{ is prime}\}$$

$$C := \{c \in \mathbb{N} : c \text{ is composite}\}$$

$$\mathbb{Z}_{>0} := \{n \in \mathbb{Z} : n > 0\}$$

$$\mathbb{Z}_{>2} := \{n \in \mathbb{Z} : n > 2\}.$$

$$p \in P$$

$$c_1, c_2, c_3 \in C$$

$$A, B, C \in \mathbb{Z}_{>0}$$

$$x, y, z \in \mathbb{Z}_{>2}$$

$$A^x = (p * c_1)^x$$

$$B^y = (p * c_2)^y$$

$$C^z = (p * c_3)^z$$

$$A^x + B^y = C^z \qquad (1)$$

From equation (1), we obtain

$$(p * c_1)^x + (p * c_2)^y = (p * c_3)^z \qquad (2)$$